Bayesian Hill Mixture Models for Marketing Mix Modeling: Synthetic Benchmarks, Component Resolvability, and Real-Data Evaluation

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Abstract

Marketing Mix Models (MMMs) typically assume that all consumers respond to advertising with a single Hill saturation curve, but real audiences are heterogeneous. We study a Bayesian mixture of Hill curves that allows multiple latent response segments to coexist within a channel, implemented in NumPyro with label-invariant convergence diagnostics and post-hoc relabeling. Three complementary experiments are reported. First, a controlled synthetic benchmark over single-Hill and mixture data-generating processes shows that mixture models improve expected log pointwise predictive density (ELPD-LOO) whenever the truth is heterogeneous and incur only a small penalty otherwise. Second, a component resolvability study varies the cosine distance between true Hill curves and characterizes how much component separation is required before the posterior recovers the underlying number of components. Third, a real-data benchmark on three Conjura organisations spanning three verticals (beauty & fitness, home & garden, toys & hobbies) shows that the three-component mixture improves ELPD-LOO over single-Hill in every organisation (by 222 to 308 units). At the same time, the strict label-invariant and relabeled convergence criteria that the mixture models satisfy on synthetic data fail uniformly for the three-component model on real data, exposing a diagnostic gap between idealised benchmarks and field data that we believe is methodologically important to report. All code, configurations, and per-case artifacts are released for reproduction.

Keywords: Marketing Mix Modeling, Bayesian inference, Hill function, mixture models, label switching, component resolvability

1 Introduction

Marketing Mix Modeling enables organisations to quantify the effectiveness of advertising spend and to optimise budget allocation. Modern implementations rely on two standard transformations: geometric or Weibull adstock to model carryover effects, and Hill saturation to model diminishing returns (Jin et al., 2017; Chan and Perry, 2017). We refer to this canonical formulation—one Hill curve per channel, fit on aggregate time-series data—as *Standard MMM*.

Standard MMM assumes that all consumers respond to a given channel with the same saturation curve. In practice, audiences are heterogeneous: heavy versus light buyers, loyalists versus switchers, and geographically distinct segments exhibit qualitatively different response patterns (Wedel and Kamakura, 2000; Allenby and Rossi, 1998). A single aggregate curve is a weighted average of these latent responses, and aggregate-level estimates can be misleading even when in-sample fit looks good. Dew et al. (2024) further argue that nonlinear and time-varying effects in MMM are jointly not identifiable from typical observational data, so apparent saturation may even be an artefact of misspecified dynamics. This identification critique is not the focus of the present paper, but it motivates studying mixture-of-Hill models as a controlled benchmark in which heterogeneity, rather than dynamics, drives the nonlinearity.

This paper studies a Bayesian mixture of Hill curves as a principled extension of Standard MMM. The model places K latent saturation curves on a single channel, with Dirichlet mixture weights, and shares adstock and baseline terms across components. We implement it in NumPyro (Phan et al., 2019) with JAX acceleration. Mixture models are well-known to suffer from label switching (Stephens, 2000; Frühwirth-Schnatter, 2006), which makes standard convergence diagnostics unreliable (Vehtari et al., 2021). We address this with three components: an ordering constraint on saturation half-points, post-hoc relabeling of MCMC draws, and label-invariant diagnostics computed on the joint log-likelihood and on scalar parameters that do not transform under label permutations.

We make three contributions. (1) A clean synthetic benchmark comparing single-Hill and mixture models under known ground truth, with predictive, recovery, and convergence metrics. (2) A *component resolvability* study that systematically varies the cosine distance between true Hill curves and traces the recovered Shannon effective component count as a function of true separation. This experiment characterises empirically how much heterogeneity must be present before a Bayesian mixture can recover the underlying segments. (3) A real-data benchmark on three Conjura organisations in three distinct verticals. The real-data results show that mixture models improve predictive density uniformly, but that the strict convergence guarantees obtained on synthetic data degrade markedly under field conditions—a diagnostic gap that we report rather than paper over.

2 Model Specification

Let x_t denote aggregated marketing spend at time t and y_t the observed outcome (purchases, revenue, or a related target). Figure 1 shows the model in plate notation.

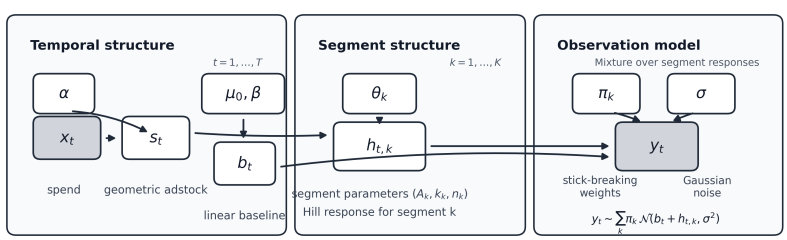

Figure 1: Graphical model. Observed variables are shaded; plates indicate replication over time steps t and mixture components k.

Adstock transformation. Geometric decay captures carryover from previous periods:

$$s_t = x_t + \alpha \cdot s_{t-1}, \quad s_0 = 0, \quad \alpha \sim \mathrm{Beta}(2, 2). \tag{1}$$

Hill saturation. Each mixture component $k \in \{1, \dots, K\}$ has its own Hill response:

$$f_k(s) = A_k \cdot \frac{s^{n_k}}{\lambda_k^{n_k} + s^{n_k}}, \tag{2}$$

with maximum effect A_k, half-saturation point λ_k, and slope $n_k \sim \mathrm{LogNormal}(\log 1.5, 0.4)$.

Mixture likelihood. The observation model is a Gaussian mixture over components:

$$y_t \sim \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}\left(\mu_0 + \beta t + f_k(s_t),\, \sigma^2\right), \tag{3}$$

where $\pi \sim \mathrm{Dirichlet}(\mathbf{1}_K)$ and μ_0, β capture intercept and trend.

Automatic prior scaling. To accommodate diverse data scales, priors on A_k, μ_0, and σ are computed from training-data statistics: $A_k \sim \mathrm{LogNormal}(\log(0.3 \cdot \mathrm{range}(y)), 0.8)$, $\mu_0 \sim \mathcal{N}(\bar{y}, 2\sigma_y)$, $\sigma \sim \mathrm{HalfNormal}(\sigma_y)$.

2.1 Identifiability and Label Switching

A finite mixture is invariant under permutations of component labels: any reordering of $(\pi_k, A_k, \lambda_k, n_k)$ yields the same likelihood. As a result, posterior summaries computed per component are meaningless without an additional convention, and standard split-$\widehat{R}$ on raw component parameters cannot distinguish genuine non-mixing from label permutations across chains (Stephens, 2000; Frühwirth-Schnatter, 2006).

Within-MCMC ordering. We impose $\lambda_1 < \lambda_2 < \cdots < \lambda_K$ via cumulative-sum reparameterization with positive increments $\delta_j \sim \mathrm{LogNormal}(\log(s_{\max}/(K+1)), 0.7)$. This restricts the posterior to one canonical labelling but can distort posterior geometry near the boundary.

Post-hoc relabeling. We additionally implement unconstrained sampling followed by post-hoc reordering: each posterior draw is permuted so that $\lambda_1 < \lambda_2 < \cdots < \lambda_K$ (Stephens, 2000; Papastamoulis, 2014). This preserves detailed balance and exposes interpretable per-component summaries.

Label-invariant diagnostics. We compute $\widehat{R}$ and effective sample sizes on three groups of quantities (Vehtari et al., 2021):

1. *Joint log-likelihood* $\sum_t \log \sum_k \pi_k \phi(y_t \mid \mu_t^{(k)}, \sigma)$, which is label-invariant by construction.
2. *Scalar parameters* (α, σ, μ_0, β) that are not affected by label permutations.
3. *Relabeled component parameters*, after the post-hoc reordering above.

A model is treated as effectively converged when all three pass: this is the criterion we use to report a "publication-ready" fit.

3 Synthetic Benchmark

3.1 Data-Generating Processes

We evaluate the models across three data-generating processes (DGPs): (1) *Single* ($K = 1$), a single Hill response; (2) *Mixture* $K = 2$, a two-component mixture; and (3) *Mixture* $K = 3$, a three-component mixture. Each DGP uses $T = 200$ observations with a 75/25 train/test split. We run five random seeds per condition.

3.2 Inference Configuration

Inference uses the No-U-Turn Sampler (NUTS) with model-specific warmup and sampling lengths and a stricter target acceptance probability for mixtures: 1,000 warmup and 1,000 samples per chain for single-Hill, 1,400 and 1,000 for $K = 2$, and 2,200 and 1,800 for $K = 3$, each with two chains. Holdout predictions inherit the adstock state accumulated during training and are evaluated at their absolute time index rather than reset to zero.

3.3 Results

Convergence. Under our combined criterion (HMC diagnostics, label-invariant $\widehat{R}$, and relabeled $\widehat{R}$), all $3 \times 3 \times 5 = 45$ synthetic fits passed the effective convergence threshold (Figure 2). Standard single-Hill diagnostics agreed with the mixture-specific label-invariant diagnostics on the synthetic data.

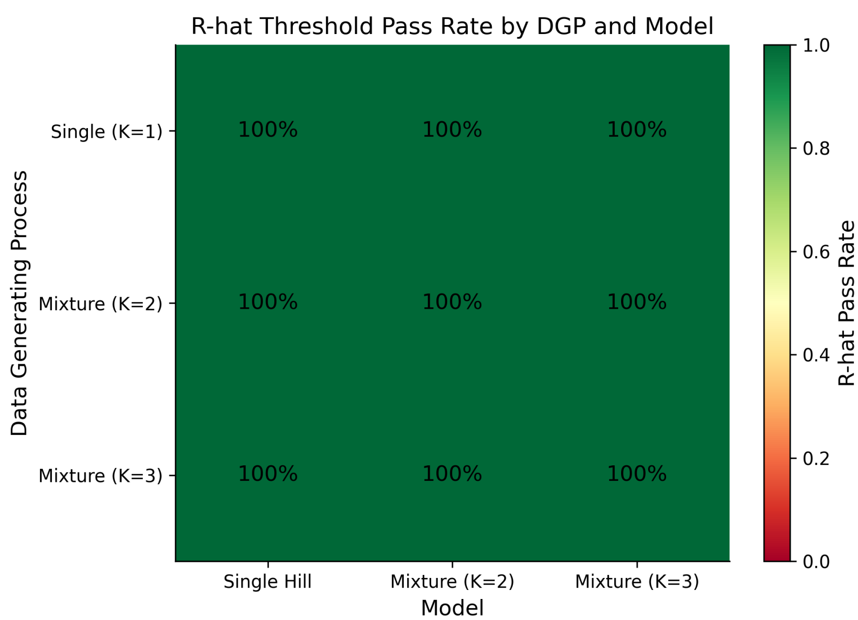

Figure 2: Effective convergence pass rate by DGP and model on the synthetic benchmark. All cells reach 100%.

Predictive performance. Figures 3 and 4 report ELPD-LOO and holdout CRPS. When the true DGP is homogeneous ($K = 1$), the three models tie within standard error. When heterogeneity is present, mixtures gain substantially in ELPD ($\Delta \approx +50$ on $K = 2$, $\Delta \approx +45$ on $K = 3$). CRPS gains are noticeably smaller than ELPD gains ($\Delta \approx 0.1$ on $K = 2$, $\Delta \approx 0.4$ on $K = 3$), indicating that the bulk of the improvement is in probabilistic density rather than in point prediction.

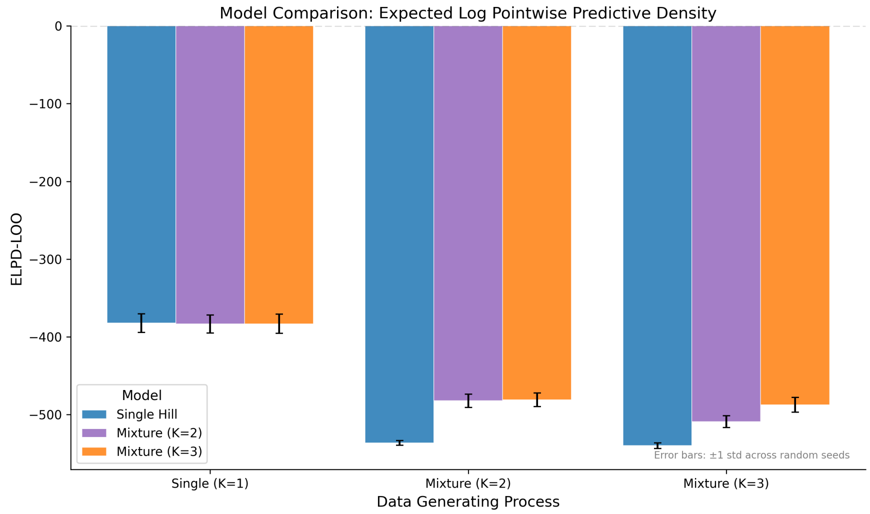

Figure 3: ELPD-LOO by DGP and model on the synthetic benchmark. Higher is better. Mixtures improve substantially when $K \geq 2$.

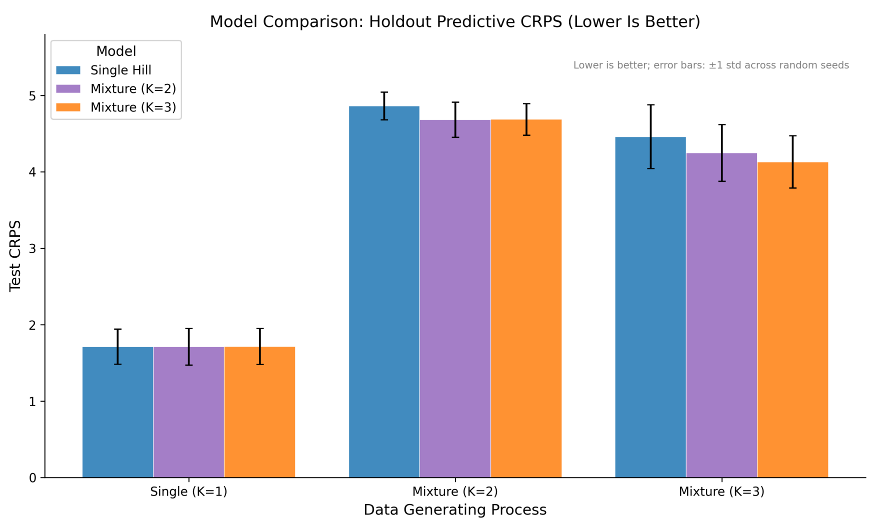

Figure 4: Holdout CRPS by DGP and model on the synthetic benchmark. Lower is better. Gains are smaller in magnitude than ELPD gains.

Latent recovery. Figure 5 shows latent-mean nRMSE on the test split. The pattern mirrors ELPD: mixtures roughly halve nRMSE under $K = 2$ ($0.14 \to 0.07$) and reduce it by a factor of ≈ 2.4 under $K = 3$ when the model is matched to the true K ($0.085 \to 0.035$). This shows that the ELPD gains correspond to genuinely better recovery of the latent response, not just better marginal calibration.

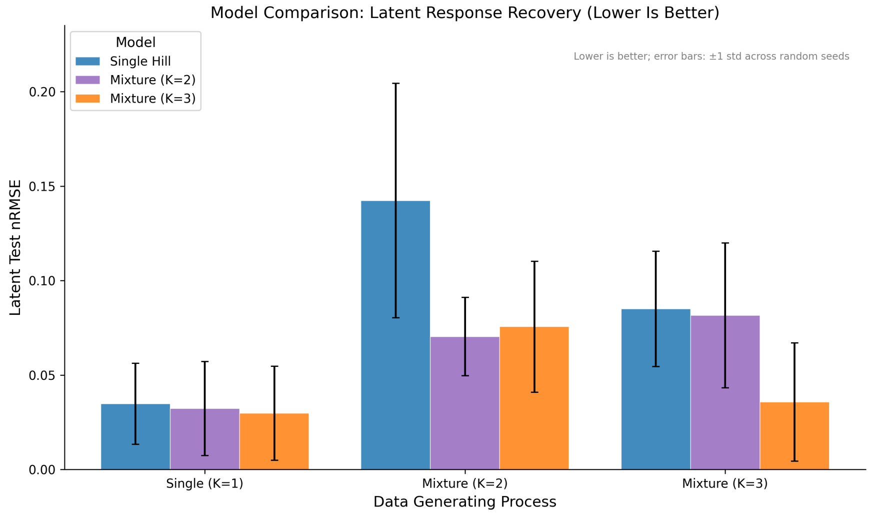

Figure 5: Latent-mean nRMSE by DGP and model. Lower is better. Mixtures recover the latent response more accurately when heterogeneity is present.

4 Component Resolvability

The previous section shows that mixtures help when heterogeneity exists, but does not tell us *how much* heterogeneity is required for the posterior to actually separate components. We address this with a controlled study.

4.1 Design

We construct nine DGP profiles by varying the spacing δ between true Hill half-points around a fixed centre c, holding amplitude and slope fixed. For $K_{\text{true}} = 1$ we use one anchor profile; for $K_{\text{true}} \in \{2, 3\}$ we use four profiles each, spanning low to extreme separation. Each profile is run with five seeds and fit with both Mixture $K = 2$ and Mixture $K = 3$, for $9 \times 2 \times 5 = 90$ fits in total.

Separation axis. For each component k we evaluate the Hill response curve on a fixed grid, compute the incremental mass m_k, and report the mean pairwise cosine distance

$$\bar{d}_{\cos} = \binom{K}{2}^{-1} \sum_{i<j} \left(1 - \frac{m_i \cdot m_j}{\|m_i\| \, \|m_j\|}\right). \tag{4}$$

Recovery axis. We summarise posterior mixture weights $\hat{\pi}_k$ by the Shannon effective count (Hill number $q = 1$) (Hill, 1973):

$$^1D = \exp\Big(-\textstyle\sum_k \hat{\pi}_k \log \hat{\pi}_k\Big). \tag{5}$$

4.2 Results

Figure 6 plots ^1D against $\bar{d}_{\cos}$ across all 90 fits. Three regimes are visible:

- For $\bar{d}_{\cos} \lesssim 0.05$, both models collapse toward the trivial value: even when $K_{\text{true}} \geq 2$, the posterior cannot resolve more than ≈ 1.5–1.7 effective components.

- Between $\bar{d}_{\cos} \approx 0.05$ and 0.3, the recovered count rises sharply with separation: this is the resolvable transition region.

- For $\bar{d}_{\cos} \gtrsim 0.3$, the recovered count saturates near K_{true}. Mixture $K = 2$ saturates near 2 by construction; Mixture $K = 3$ tracks the true K and approaches 3 on $K = 3$ DGPs.

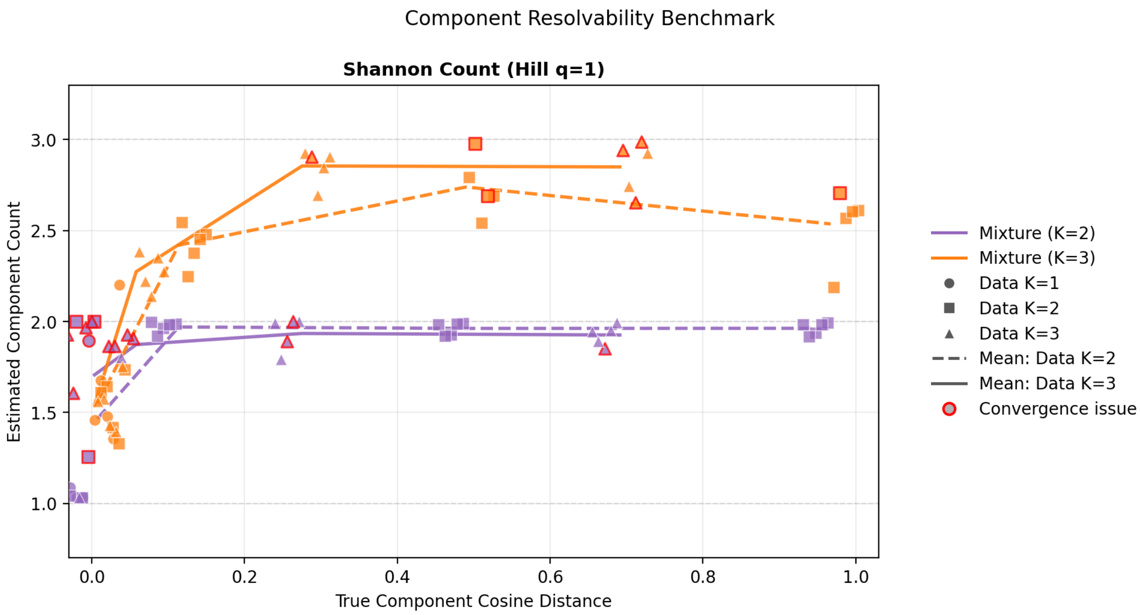

Figure 6: Component resolvability: Shannon effective count of the posterior versus mean pairwise cosine distance of the true Hill curves. Red-outlined points indicate cases that failed the publication-ready convergence criterion.

Failure modes. The publication-ready convergence criterion fails most frequently in two regimes. First, under-specified fits at low separation: Mixture $K = 2$ on $K_{\text{true}} = 3$ DGPs at low δ shows publication-fail rates of 0.6–0.8. Second, at the highest separation under $K_{\text{true}} = 3$, Mixture $K = 3$ exhibits a non-trivial failure rate (0.6 at the most extreme profile), suggesting that the strict diagnostic criterion is sensitive to posterior geometry even when the components are far apart.

Practical implication. Combined with the ELPD results of Section 3, the resolvability picture suggests that mixture-of-Hill models can be used reliably when the cosine separation between true components exceeds ≈ 0.1. Below this threshold, the posterior is consistent with a smaller effective K, and the modeller should expect both lower convergence pass rates and very limited information gain over a single-Hill baseline.

5 Real-Data Benchmark

5.1 Dataset and Organisation Selection

We use the public Conjura MMM dataset, which contains daily spend across nine advertising channels and several purchase-related target variables for a panel of e-commerce organisations.

Following the per-organisation modelling convention of MMM practice, we treat each organisation as an independent time series and do not share parameters across organisations.

To select organisations for the benchmark, we screen all candidates with at least 200 daily observations and at least two active spend channels (non-zero on more than 10% of days), and rank them by a composite score combining series length, spend coefficient of variation, spend–target correlation, and channel breadth. We pick three top-ranked organisations from three different verticals to support a modest cross-vertical claim. The selected organisations are summarised in Table 1.

Table 1: Selected Conjura organisations for the real-data benchmark.

Vertical	T (days)	Active channels	Spend–target r	Total spend (USD M)
Toys & Hobbies	1,375	5	0.95	3.3
Beauty & Fitness	924	8	0.96	5.2
Home & Garden	1,642	5	0.80	2.1

5.2 Inference Configuration

Each organisation is fit with three models—single-Hill, Mixture $K = 2$, and Mixture $K = 3$—under three seeds, giving 27 fits in total. Warmup and sampling are set to $(600, 600)$, $(900, 900)$, and (1,600, 1,200) for single-Hill, $K = 2$, and $K = 3$ respectively, with two chains each and target acceptance increased for larger mixtures. The 75/25 train/test split preserves adstock state and absolute time index as in the synthetic benchmark.

5.3 Predictive Performance

Table 2 reports ELPD-LOO, holdout MAPE, holdout CRPS, and holdout 90% coverage averaged across seeds. The key observation is that *Mixture $K = 3$ achieves the best ELPD-LOO in every organisation*, with improvements of +222 (beauty & fitness), +308 (home & garden), and +275 (toys & hobbies) over the single-Hill baseline. Mixture $K = 2$ is intermediate.

Table 2: Real-data predictive performance, averaged across three seeds. ELPD-LOO higher is better; MAPE/CRPS lower is better; 90% coverage closer to 0.9 is better.

Organisation	Model	ELPD-LOO	Test MAPE	Test CRPS	90% cov.
Beauty & Fitness	Single Hill	−3,008	30.3	108	0.33
	Mixture $K = 2$	−2,992	29.6	101	0.37
	Mixture $K = 3$	**−2,786**	34.3	122	0.54
Home & Garden	Single Hill	−6,168	26.0	25.0	0.87
	Mixture $K = 2$	−5,985	31.4	27.2	0.88
	Mixture $K = 3$	**−5,860**	28.2	25.9	0.86
Toys & Hobbies	Single Hill	−6,379	36.4	57.3	0.94
	Mixture $K = 2$	−6,346	29.3	53.5	0.98
	Mixture $K = 3$	**−6,104**	34.3	47.3	0.96

Figure 7 visualises the same comparison. The Mixture $K = 3$ advantage is consistent in sign across organisations, while the absolute scale of ELPD-LOO varies by an order of magnitude with series length and target variance.

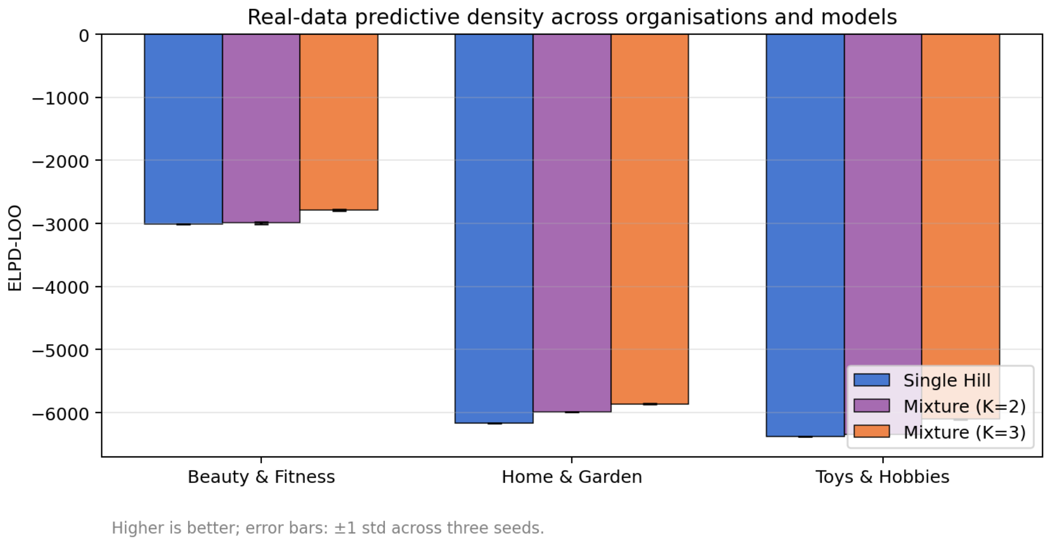

Figure 7: Real-data ELPD-LOO across the three organisations and three models. Higher is better. Error bars are ± 1 standard deviation across three random seeds. Mixture $K = 3$ achieves the highest ELPD-LOO in every organisation, with seed-level variability much smaller than the model-level gap.

The relationship between ELPD and the other metrics is heterogeneous. On *home & garden* the rank ordering of models is consistent across ELPD, CRPS, and MAPE: the mixture improves all three. On *toys & hobbies*, ELPD and CRPS both improve, while MAPE is non-monotonic. On *beauty & fitness*, ELPD improves substantially but MAPE and CRPS deteriorate slightly for $K = 3$, and the model remains visibly underconfident: 90% coverage rises from 0.33 (single) to 0.54 ($K = 3$) but stays well below the nominal level, indicating that the predictive intervals are too narrow even for the best fit. We return to this in the discussion.

To make concrete what the mixture is doing, Figure 8 shows the posterior response curves for the home & garden organisation under all three models on a single representative seed. Single-Hill recovers one slowly-saturating curve. Mixture $K = 2$ resolves two components with weights 0.93 and 0.07, in which the small minority component carries a much higher saturation ceiling. Mixture $K = 3$ uses two active components with weights 0.84 and 0.14 and a third component that collapses to weight 0.02 (labelled *inactive* in the legend). The qualitative two-mode structure is robust to whether K is set to 2 or 3, which is consistent with the posterior effective component count reported below.

5.4 Convergence on Real Data

Table 3 summarises the rate at which each (organisation, model) cell passes the publication-ready convergence criterion (HMC diagnostics plus label-invariant and relabeled $\widehat{R}$) across the three seeds. Two patterns are stark.

First, *Mixture* $K = 3$ *passes the publication-ready criterion in* 0/9 *fits* on real data, despite passing in 100% of the synthetic fits at matched K. Second, the posterior mean effective K saturates near 2 for all three organisations: even Mixture $K = 3$ recovers no more than ≈ 2.5

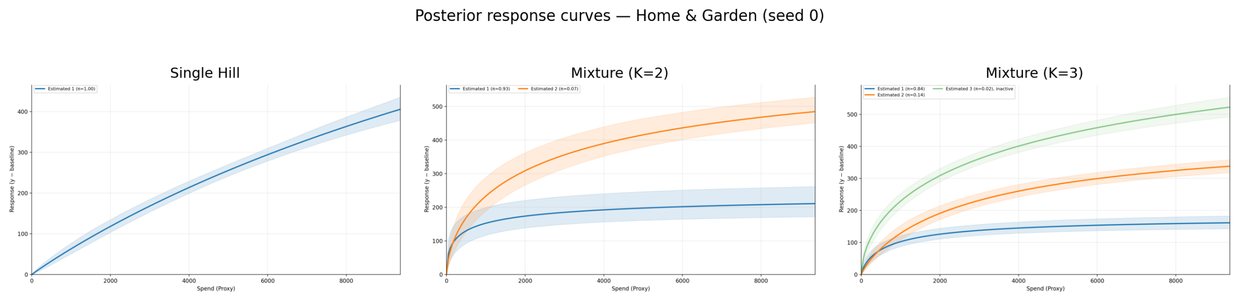

Figure 8: Posterior response curves for Home & Garden (seed 0) under Single Hill, Mixture $K = 2$, and Mixture $K = 3$. Shaded bands are 90% posterior intervals; legend entries report posterior mixture weights $\hat{\pi}$ per component. The Mixture $K = 3$ third component is labelled *inactive* because its weight collapses to 0.02.

Table 3: Real-data convergence pass rates (publication-ready criterion across three seeds), posterior mean effective K, and mean number of LOO observations with Pareto $k > 0.7$.

Organisation	Model	Pub. pass rate	Effective K	Pareto k bad (mean)
Beauty & Fitness	Single Hill	0.67	1.00	1.7
	Mixture $K = 2$	0.00	1.65	15.0
	Mixture $K = 3$	0.00	2.17	3.3
Home & Garden	Single Hill	0.67	1.00	0.7
	Mixture $K = 2$	0.33	1.94	0.7
	Mixture $K = 3$	0.00	2.00	1.7
Toys & Hobbies	Single Hill	1.00	1.00	2.7
	Mixture $K = 2$	0.67	2.00	1.7
	Mixture $K = 3$	0.00	2.45	3.3

effective components. The mixture model does extract additional structure relative to a single curve, but the data do not warrant three distinct components under our diagnostic criterion.

The high Pareto-k count for the beauty & fitness Mixture $K = 2$ fit (15 observations on average) is consistent with a moderately misspecified model on data with influential outliers, and points to robustifying the likelihood as a useful next step.

6 Discussion

Mixture helps, but mainly in predictive density. Across both synthetic and real data, the most consistent benefit of mixture-of-Hill over single-Hill is in ELPD-LOO. The CRPS gains are smaller, and improvements in MAPE are organisation-specific. This pattern—predictive density improves first—is consistent with the view that mixtures primarily improve probabilistic calibration and uncertainty representation rather than mean prediction. A practitioner whose objective is point forecasting may gain little; a practitioner who needs calibrated predictive intervals for budget decisions gains more.

The synthetic–real diagnostic gap. The publication-ready convergence criterion that passes on 100% of synthetic fits fails on 0/9 Mixture $K = 3$ fits on real data. This is not a contradiction: synthetic data are generated within the model class, with controlled noise and stationary mixing weights. Real organisations exhibit additional structure—seasonality, regime shifts, channel-mix changes, and outliers—that the within-model strict criterion is sensitive to. We report this gap rather than relaxing the criterion, because the gap itself is one of the most informative findings: it tells practitioners that synthetic-benchmark convergence does not automatically transfer.

Effective K saturates near 2. Across all three organisations, even the $K = 3$ model recovers no more than ≈ 2.5 effective components in the posterior. Combined with the resolvability evidence that components are only identifiable when the true cosine separation exceeds ≈ 0.1, this is best read as a soft upper bound for this kind of dataset: with weekly-to-daily aggregated spend at the organisation level, two latent saturation modes are about as much structure as the data can support.

Beauty & Fitness as a cautionary case. The Beauty & Fitness organisation has the strongest mixture-versus-single ELPD gap but also the most striking calibration failure: 90% holdout coverage rises from 0.33 to 0.54 across models but never approaches the nominal level. The high Pareto-k count further indicates influential observations. The honest reading is that the mixture is doing real work but is still not a sufficient model for this organisation, and that a heavier-tailed likelihood or richer observation model would be needed before deploying it for decisions.

Identifiability caveat. As Dew et al. (2024) note, nonlinear and time-varying effects in MMM are jointly under-identified from typical observational time series. Our mixture-of-Hill model does not resolve this concern; it offers a controlled extension of the standard family in which the source of nonlinearity (heterogeneity, in this case) is at least made explicit. Stronger causal claims would require external variation, such as geo-experiments or incrementality tests.

Limitations. We aggregate spend across channels into a single x_t in both benchmarks, which simplifies inference but loses cross-channel information. The model assumes constant mixing weights over time. The real-data evaluation covers three organisations and three verticals; cross-organisation generalisation claims would require a substantially larger panel. Finally, the prior on λ depends on the empirical spend range; this is convenient but couples prior to data in a way that should be made explicit in any deployment.

7 Conclusion

We presented a Bayesian mixture of Hill saturation functions for Marketing Mix Modeling, together with a label-invariant diagnostic workflow and three complementary benchmarks: a synthetic comparison under known ground truth, a component-resolvability study quantifying when components can be separated, and a real-data evaluation across three organisations in distinct verticals. Mixture-of-Hill consistently improves ELPD-LOO under heterogeneity, both synthetic and real, with mean improvements of +222 to +308 over single-Hill on the real organisations. The improvements appear primarily in predictive density rather than point prediction. The most important caveat is methodological: the strict publication-ready convergence criterion that succeeds uniformly on synthetic data fails uniformly for Mixture $K = 3$ on real data, and the posterior effective K saturates near 2. We interpret this gap as a directly useful finding for practitioners deploying mixture MMMs, not as a weakness to be sanded away. Code and per-case artifacts are available at `https://github.com/jpcca/marketing-mix-modelling`.

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